Wiring Reference

I2C Bus (Shared by all 3 sensors)

All three sensors share a single I2C bus on the Pico W.

Signal	Pico W Pin	GPIO	Wire Color
SDA	Pin 1	GP0	Orange
SCL	Pin 2	GP1	Yellow
3V3	Pin 36	—	Red
GND	Pin 38	—	Black

Run a dedicated row on the breadboard for each signal so all sensors tap into the same rail.

SHT31 (Temperature + Humidity) — I2C address `0x44`

SHT31 Pin	Connect to
VIN	3V3 rail
GND	GND rail
SCL	SCL rail (GP1)
SDA	SDA rail (GP0)

TSL2591 (Light / Lux) — I2C address `0x29`

TSL2591 Pin	Connect to
VIN	3V3 rail
GND	GND rail
SCL	SCL rail (GP1)
SDA	SDA rail (GP0)
3Vo	Leave unconnected
INT	Leave unconnected

BMP390 (Pressure + Temperature) — I2C address `0x77`

BMP390 Pin	Connect to	Notes
VIN	3V3 rail	

GND	GND rail	
SCK	SCL rail (GP1)	Adafruit labels clock as SCK
SDI	SDA rail (GP0)	Adafruit labels data as SDI
CS	Leave unconnected	Floating = I2C mode. Do NOT tie to 3V3
SDO	Leave unconnected	Floating = address 0x77. Do NOT tie to 3V3
3Vo	Leave unconnected	
INT	Leave unconnected	

> ***Important:*** *Tying CS or SDO to 3V3 puts the chip in SPI mode and it will not appear on the I2C bus.*

SD Card Adapter (SPI) — to be wired

The SD card uses SPI, a separate bus from I2C. Suggested pin assignments:

SD Adapter Pin	Pico W Pin	GPIO
VCC / 3V3	Pin 36	—
GND	Pin 38	—
MOSI	Pin 5	GP3
MISO	Pin 6	GP4
SCK	Pin 7	GP5
CS	Pin 9	GP6

I2C Address Summary

Sensor	Address
TSL2591	`0x29`
SHT31	`0x44`
BMP390	`0x77`